

Adaptive Trial Designs

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Introduction

Adaptive designs allow pre-specified modifications – sample size, randomisation ratios, treatment-arm selection – based on data accrued during the trial. They promise efficiency but require careful statistical control to preserve Type I error. The FDA and EMA both provide detailed guidance on acceptable adaptive modifications.

Prerequisites

Group-sequential trials; alpha spending; sample-size calculation.

Theory

Common adaptive features: - **Sample-size re-estimation** (blinded or unblinded). - **Early stopping** for efficacy or futility. - **Arm selection** (drop inferior arms in multi-arm trials). - **Response-adaptive randomisation** (shift allocation toward effective arms). - **Population enrichment** (restrict to a responsive subgroup).

Maintaining Type I error under pre-specified adaptations requires methods like group-sequential boundaries, combination tests, or conditional error functions.

Assumptions

Adaptation rules are fully pre-specified (protocol, SAP); unblinded information access is tightly controlled (Data Monitoring Committee); adjustments are statistically valid.

R Implementation

```
library(rpact)

# Group-sequential design with O'Brien-Fleming boundary, 3 analyses
design <- getDesignGroupSequential(
  sided = 2, alpha = 0.025, beta = 0.2,
  typeOfDesign = "OF",
  informationRates = c(0.5, 0.75, 1)
)
kable_summary <- summary(design)
print(design)

# Plan sample size for a two-arm trial with continuous outcome
ssr <- getSampleSizeMeans(design = design,
  alternative = 0.3, stDev = 1)
print(ssr)
```

Output & Results

Group-sequential boundaries and associated sample sizes per stage; cumulative Type I error preserved at alpha.

Interpretation

“The adaptive design applied O’Brien-Fleming boundaries at 50 %, 75 %, and 100 % information; stage-1 interim inefficacy boundary would stop at $p > 0.0002$, preserving overall $\alpha = 0.025$.”

Practical Tips

- Pre-specify every adaptation (including the decision rule) in the protocol.
- FDA/EMA require a detailed justification of the adaptive feature and its operating characteristics.
- Independent data monitoring committee is essential for efficacy or futility stopping.
- Simulation is often used to verify operating characteristics; report them.
- Post-hoc adaptations (“seamless” trial extensions) are exploratory, not confirmatory.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale